

COAL Final Examination

Important topics:

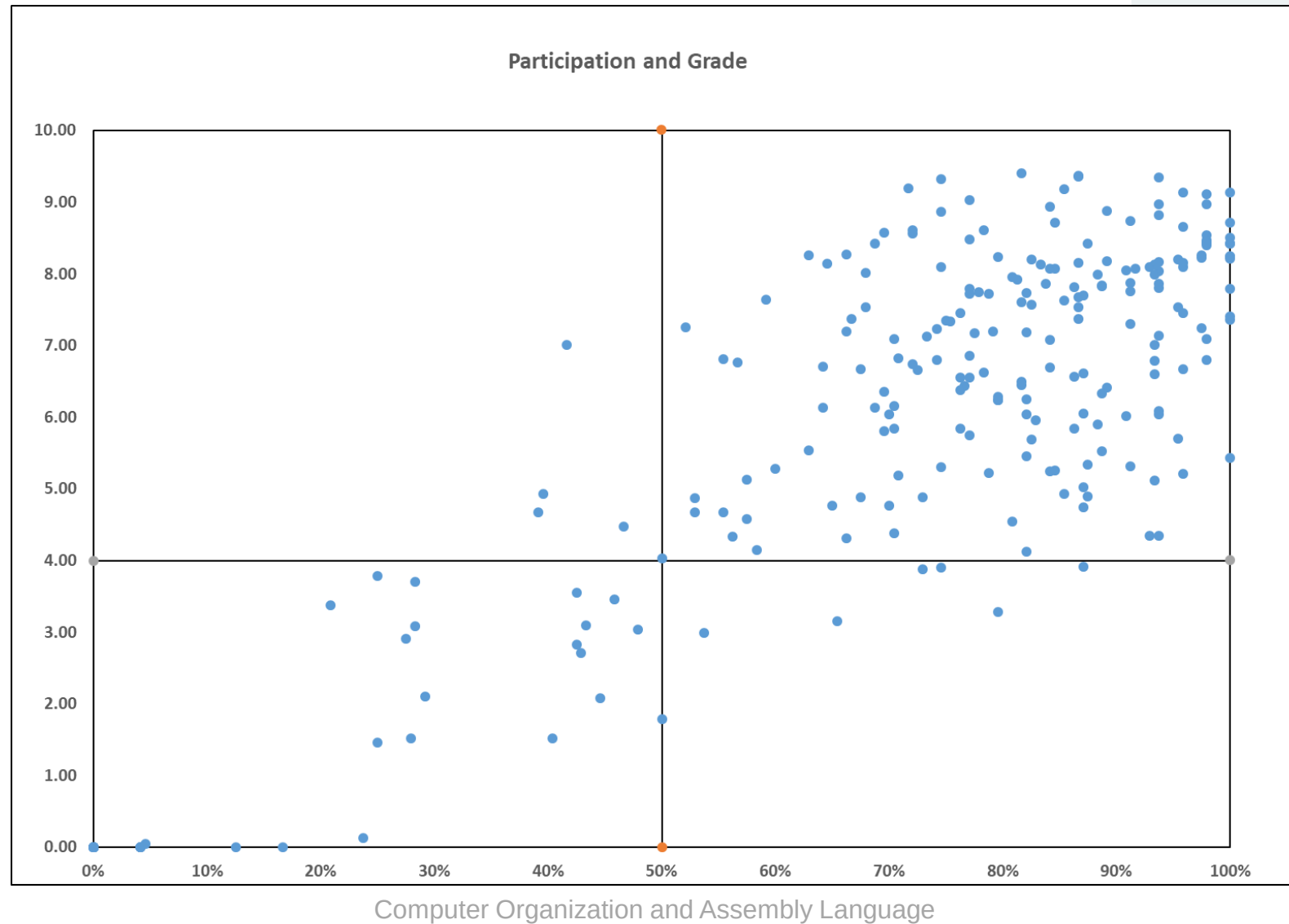
- Hexadecimal representation of integer numeric values in registers and memory
- Execution of simple instructions including arithmetic calculations
- Loop instruction, nested loops
- String processing
- Procedures, instructions CALL and RET, parameter passing
- Interrupts, interrupt vectors, hardware and software interrupts
- Timer interrupt
- Programmable Interval Timer and sound
- Video adapter basics (text mode and graphic mode)
- Keyboard basics, keyboard interrupt

Examples included in the exam could be based on variations of code examples discussed in lectures

Assessment (see COAL Requirements)

- At the end of the course all individual practice assignments (labs) are evaluated with one common grade (see Requirements document on the course page for details)
- If less than 4 individual assignments (labs) have been completed, then the average grade for all practice assignments is 0 (zero)
- Average grades of tests and of assignments are calculated before the first exam and cannot be changed later
- The final grade of the study course is calculated according to formula:
final grade = 0.25 x tests + 0.45 x assignments + 0.30 x exam
In the case the student receives non-passing exam mark (i.e. less than 4) the exam mark is used as the final grade (the formula is not applied)
- If the calculated final grade is less than 4, the exam must be re-taken in accordance with RTU regulation

Participation and final grade forecast (as of 09/12/2025)



Example

Integer numeric value N in doubleword format is defined.

N dword 267

Write internal hexadecimal code of each of the bytes starting from the address N.

ANSWER (hexadecimal value):

N	N+1	N+2	N+3
0B	01	00	00

$$267 = 256 + 11 = 100h + 0Bh = 10Bh$$

Example

Array of five integer numeric values (in word format) is defined. Each of the five values are stored in two bytes. Address of the first element of the array is N. Write internal hexadecimal code of the memory byte with address N+4.

N word 1, 10, -10, 12, -15

Write content of the memory byte N+4 (two hexadecimal digits without any other symbols).

ANSWER: **F6**

Explanation:

N	N+1	N+2	N+3	N+4	N+5	N+6	N+7	N+8	N+9
01	00	0A	00	F6	FF	0C	00	F1	FF

Example

General purpose register EBX contains internal code (hexadecimal):

010203F9

What the value of the register BX will be after execution of the following instructions?

```
inc    ebx
mov    bh, 0FFh
add    bx, 7
```

010203FA

0102FFFA BX = FFFA = -6

01020001 BX = -6 + 7 = 1

ANSWER (decimal value):

1

Example

What internal code of the memory field **R** will be after execution of the following instructions?

```
.data
val    byte    -1, 2, -3, 4, 5, -6, 7, 8 ; array given
R      word    0

.code
start  proc
        mov esi, 2
        mov ax, 0
        mov ecx, 2
B:      add al, val[esi]
        add esi, 2
        loop  B
        mov  R, ax
; .....
start endp
        end start
```

ANSWER (hexadecimal value):

R	R+1
02	00

Example

Fragment of the code to find total number of the ampersand symbols (&) in the zero terminated text given. Write missing instruction(s) in the box to complete this code.

```
.data
```

```
txt byte '012&456&891&ABC', 0 ; text string
```

```
.code
```

```
start
```

```
proc
```

```
mov dx, 0
```

```
mov edi, 0
```

```
check: cmp txt[edi], 0
```

```
je done
```

```
cmp txt[edi], '&'
```

```
jne next
```

```
inc dx
```

```
next: inc edi
```

```
jmp check
```

```
done:
```

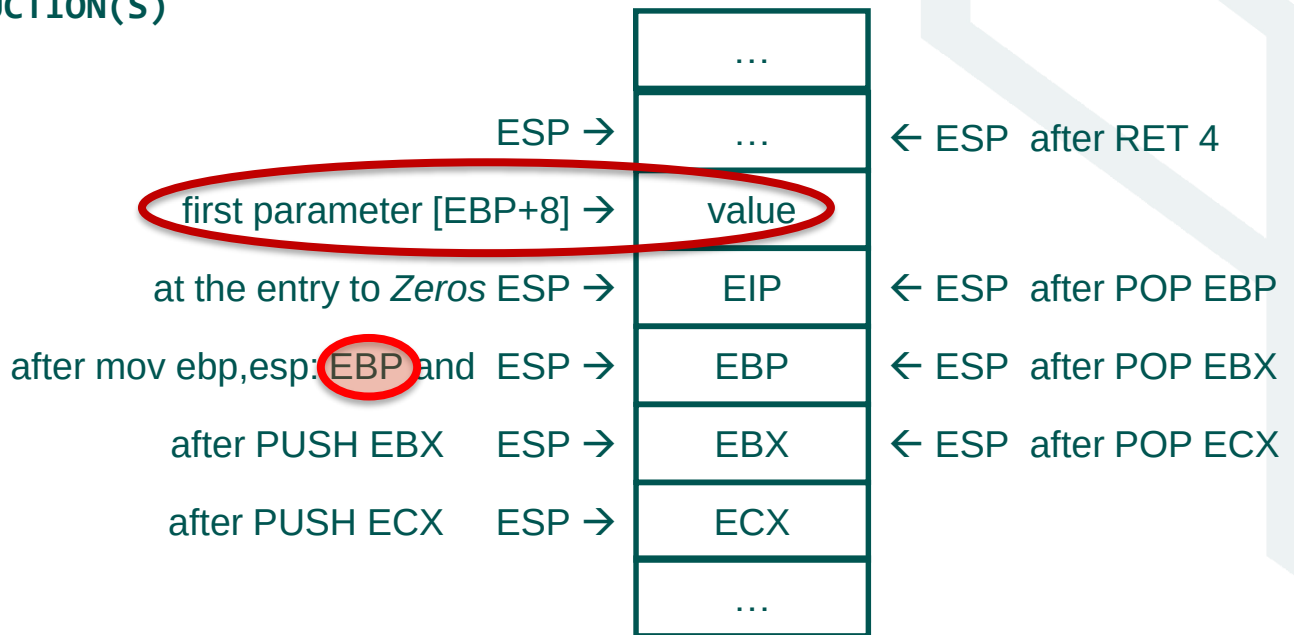
```
; number of ampersand symbols is in dx
```

Write missing instructions here

Example

Procedure Zeros receives one doubleword parameter via stack and returns in the register EAX number of bits which contain zero in this doubleword.

```
Zeros proc near
    push ebp
    mov  ebp, esp
    push ebx
    push ecx
    ; MISSING INSTRUCTION(S)
    xor  ebx, ebx
    mov  ecx, 32
t:    test eax, 1
    jnz  next
    inc  ebx
next: shr  eax, 1
    loop t
    mov  eax, ebx
    pop  ecx
    pop  ebx
    pop  ebp
    ret  4
Zeros endp
```



Which is a correct instruction to insert in the line of code marked by the comment?

- A. mov eax, [ebp+12]
- ☒ B. mov eax, [ebp+8]
- C. mov ecx, [ebp+8]
- D. mov ecx, [esp+8]
- E. None of the above

Example

Procedure Zeros receives one doubleword parameter via stack and returns in the register EAX number of bits which contain zero in this doubleword.

```
Zeros proc near stdcall
```

```
    push ebp
    mov  ebp, esp
    push ebx
    push ecx
    mov  eax, [ebp+8]
    xor  ebx, ebx
    mov  ecx, 32
t:    test eax, 1
      jnz next
      inc  ebx
next: shr  eax, 1
      loop t
    mov  eax, ebx
    pop  ecx
    pop  ebx
    pop  ebp
    ret  4
```

```
Zeros endp
```

Write instructions to call this procedure and count zero bits in the doubleword Beta defined as

```
Beta    dword    -12
```

ANSWER:

<pre>push Beta call Zeros</pre>

Example

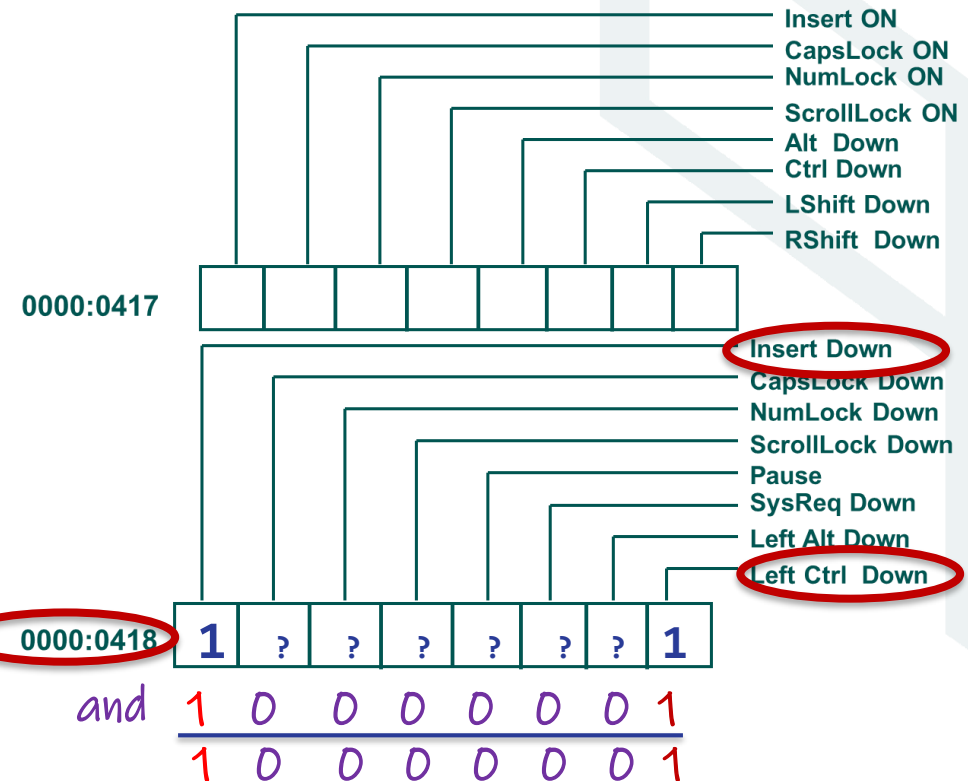
A program is given to show message "Insert and LCtrl keys are down" if Insert and LeftCtrl keys are both held down at the moment when this code is executed. Some instructions are missing. Write one or more missing instructions in the code lines marked by the comment.

```
.model small
.data
msgInit    byte    "Press any key or Esc to exit!", 13, 10, "$"
msgDown    byte    "Insert and LCtrl keys are down", 13, 10, "$"
.code
.startup
    mov     ah, 9h
    mov     dx, offset msgInit
    int     21h
    xor     bx, bx
    mov     es, bx
tst:                ; MISSING INSTRUCTIONS

    mov     dx, offset msgDown
    mov     ah, 9
    int     21h
notBothDown:
    mov     ah, 0      ;BIOS interrupt service
    int     16h        ;wait for key press
    cmp     ah, 1      ;Esc scan code?
    jne     tst
.exit
end
```

ANSWER :

```
mov al, es:0418h
and al, 10000001b
cmp al, 10000001b
jne notBothDown
```



Example

Programmable Interval Timer (PIT) is used to generate sound on computer's internal speaker. What value must be loaded to the counter register of the PIT Channel 2 in order to generate output sound of approximately 2000 Hz frequency?

The oscillator provides roughly 1193182 Hz input to each channel of the PIT. Counter register of PIT channel contains 16 bit non-signed positive value.

ANSWER (decimal value):

597

Explanation

Each input impulse from the oscillator decrements the counter register by 1. When the counter has reached 0 the output is generated and the counter is reset again to the initial value. For example, if the initial counter value is 10 the output impulse will be generated after each 10 input impulses, i.e. output frequency will be $1193182/10 \approx 119328 \text{ Hz}$ (Output = Input/Counter).

To calculate counter value for the output frequency required you may use a formula $\text{Counter} = \text{Input frequency} / \text{Output frequency}$.

In this case to generate 2000 Hz sound the counter register should contain value $1193182/2000 = 597$.

Example

Video adapter is in the VGA graphics mode 12h with 16 colors and 640x480 pixels resolution. How many bytes are used to store information in the video memory about one horizontal line of 32 red color pixels at upper left corner of the screen?

ANSWER:

16

Explanation

In the VGA graphics mode each pixel on the screen corresponds to one bit in all display planes of the video memory. High order bit of the byte A000:0000 corresponds to the pixel in the upper left corner of the screen. Each display plane memory has the same starting address A000:0000.

Combination of bits from all display planes determines the colour of the pixel. To encode 16 different colours 4 display planes are used (4 bits may store 16 different unsigned values from 0 to 15). To store the information about one pixel VGA uses one bit in each of the 4 display planes.

For 32 pixels of the horizontal line 4 bytes are used in each of the display planes (one byte for 8 pixels). So 16 bytes are used to store information about 32 pixels long horizontal line (4 bytes x 4 planes). Colour of the line does not matter.

Example

Video adapter is in the VGA text mode with 25 rows, 80 symbols per row and 16 colors for symbols. Two letters OK should be displayed at the end of the first row. Write one or more missing instructions in the code lines marked by the comment to assign corresponding value to the register AX.

```
.model small
.code
.startup

        ; MISSING INSTRUCTION(S)
        ; set address
mov bx, 0B800h
mov es, bx ; segment value for text mode video memory
mov di, dx ; offset
        ; write symbols and attributes
mov byte ptr es:[di], 'O'
mov byte ptr es:[di+1], 01110100b ; white background, red symbol
mov byte ptr es:[di+2], 'K'
mov byte ptr es:[di+3], 01110100b ; white background, red symbol
.exit
end
```

ANSWER:

mov dx, 156

Explanation: In this VGA text mode each two bytes (one word) in the video memory contains information about one symbol on the screen. Information about the first symbol in the first row is in two bytes at address B800:0000. To calculate offset value you may use formula $\text{offset} = 2 * ((\text{row} - 1) * 80 + (\text{col} - 1))$. In this case $\text{offset} = 2 * ((1 - 1) * 80 + (79 - 1)) = 156$